Shubham Sah

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Education

Indian Institute of Information Technology (IIIT) — **Bhopal, India**
Bachelor of Technology in Computer Science and Engineering (CGPA: 7.59) — *Aug. 2023 – May 2027*

Experience

RevMozi — **Bengaluru, India**
Software Development Engineer (SDE) Intern — *Jul. 2026 – Present*

- **Developed** scalable backend microservices and optimized REST APIs, enhancing database queries to improve application performance and reliability.
- **Collaborated** directly with the CTO on system architecture, implementing unit/integration test suites to ensure robust code quality.

AlgoUniversity — **Hyderabad, India**
MTS Intern (Backend & Systems Engineering) — *May 2026 – July 2026*

- **Developed** core backend microservices for the education platform, successfully handling 500+ daily active users with sub-200ms latency.
- **Automated** operational CI/CD pipelines using Python and Bash scripting to streamline internal developer workflows.
- **Implemented** robust system telemetry and logging to proactively diagnose and resolve production bottlenecks.

Indian Institute of Information Technology (IIIT) — **Bhopal, India**
Undergraduate Researcher (Quantum-Safe ZKP) — *Aug. 2025 – May 2026*

- **Designed** a lattice-based Zero-Knowledge Proof (ZKP) protocol prototype in Python to evaluate post-quantum secure verification schemes.

Projects

Remote Browser Control | ***React, Node.js, Docker, WebSockets, Puppeteer, AWS*** GitHub | Live

- **Architected** a remote browser control pipeline using Node.js, WebSockets, and Puppeteer/CDP to stream high-performance Chromium screencasts.
- **Integrated** real-time mouse coordinate normalization and keyboard key emulation on an HTML5 canvas, reducing interaction latency below 100ms.
- **Resolved** connection socket leaks and zombie container processes by enforcing strict WebSocket-to-lifespan container bindings.

Enterprise CI/CD Deployment Platform | ***GitHub Actions, Docker, Trivy, Render, Bash*** GitHub | Live

- **Architected** a reusable CI/CD platform using GitHub Actions, enabling zero-config onboarding for full-stack apps via 10 lines of YAML.
- **Engineered** multi-stage, non-root Docker builds, integrating Aqua Trivy to block CRITICAL vulnerability containers from reaching production.
- **Implemented** zero-downtime automated rollouts to Render via GHCR, utilizing Bash polling scripts to verify health and halt failed updates.

Skills & Expertise

Infrastructure & DevOps: Cloud Computing (AWS/GCP/Render), Docker, Linux/Bash Scripting, System Architecture
Security & Backend: Python, Django, C/C++, RESTful APIs, Applied Cryptography, Zero-Knowledge Proofs
Data & Databases: SQL (SQLite/MySQL), MongoDB, Data Analytics, Data-driven Decision Making
Developer Tools: Git/GitHub, VS Code, Postman, Agile Methodologies, Cross-functional Collaboration, LaTeX
Core Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems

Publications

- **Quantum-Safe ZKP Architecture for Secure Verification** — Authored a research paper detailing a lattice-based Zero-Knowledge Proof protocol; submitted to ***Cluster Computing Journal***.
- **Arduino-Based Robotic Car** — Published an IEEE paper on real-time embedded hardware control and ultrasonic sensor obstacle avoidance; ***IEEE SCEECS 2024***.

Achievements

- **Solved** 770+ algorithmic challenges across LeetCode, Codeforces (Rating 1863), and CodeChef (5-Star) to optimize complex data structures.
- **Committed** open-source code enhancements to distributed developer repositories during GirlScript Summer of Code (GSSoC).
- **Secured** top 15 rank out of 500+ students in competitive coding at IIIT Bhopal.